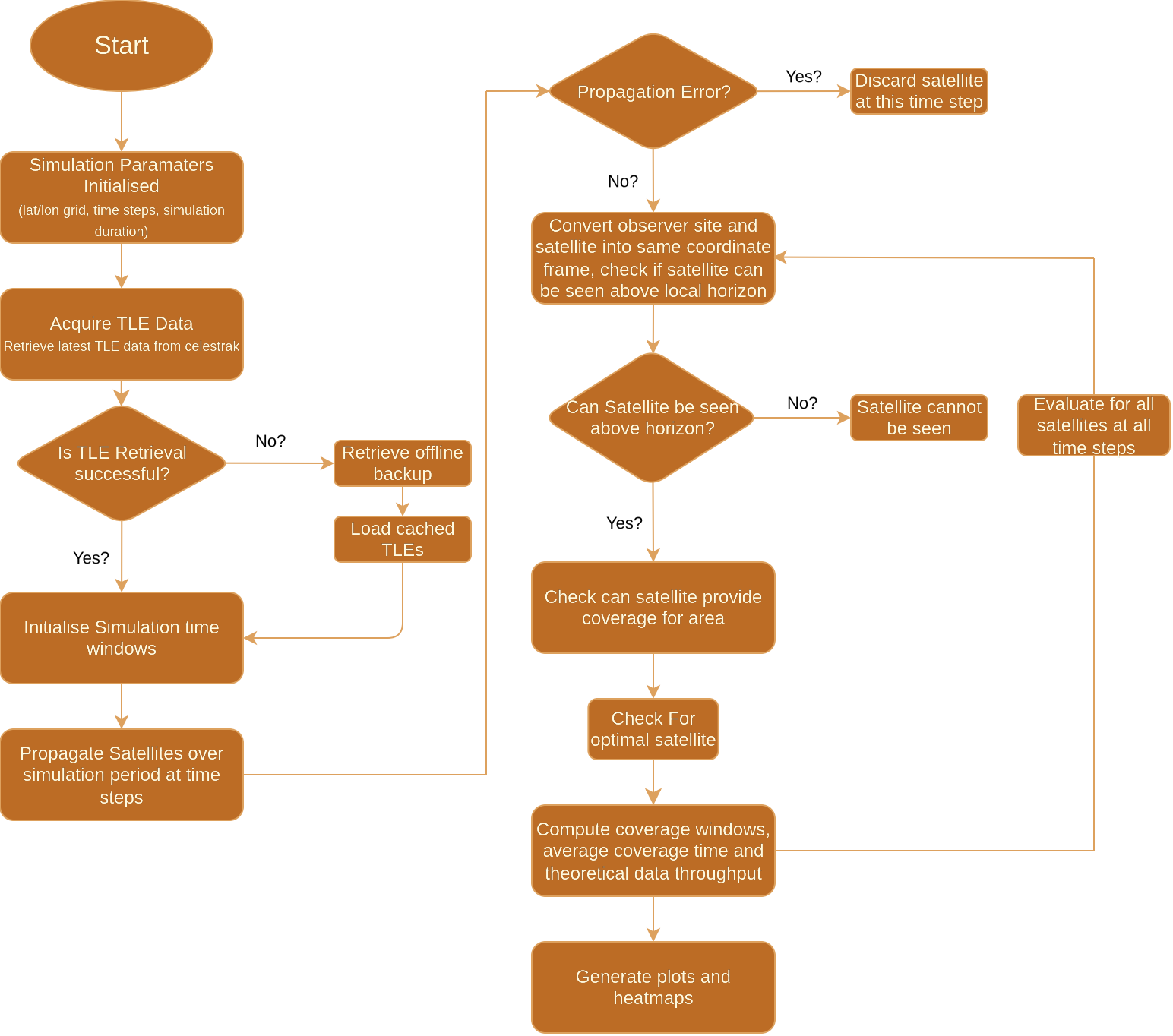
Start
Simulation Paramaters Initialised
(lat/lon grid, time steps, simulation duration)
Acquire TLE Data
Retrieve latest TLE data from celestrak
Is TLE Retrieval successful?
No?
Retrieve offline backup
Load cached TLEs
Yes?
Initialise Simulation time windows
Propagate Satellites over simulation period at time steps
Propagation Error?
Yes?
Discard satellite at this time step
No?
Convert observer site and satellite into same coordinate frame, check if satellite can be seen above local horizon
Can Satellite be seen above horizon?
No?
Satellite cannot be seen
Evaluate for all satellites at all time steps
Yes?
Check can satellite provide coverage for area
Check For optimal satellite
Compute coverage windows, average coverage time and theoretical data throughput
Generate plots and heatmaps